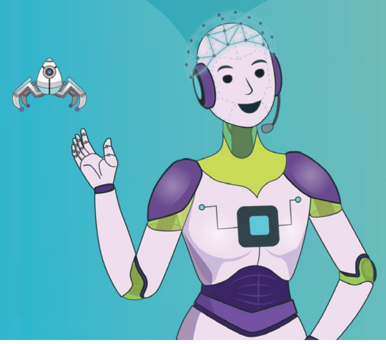
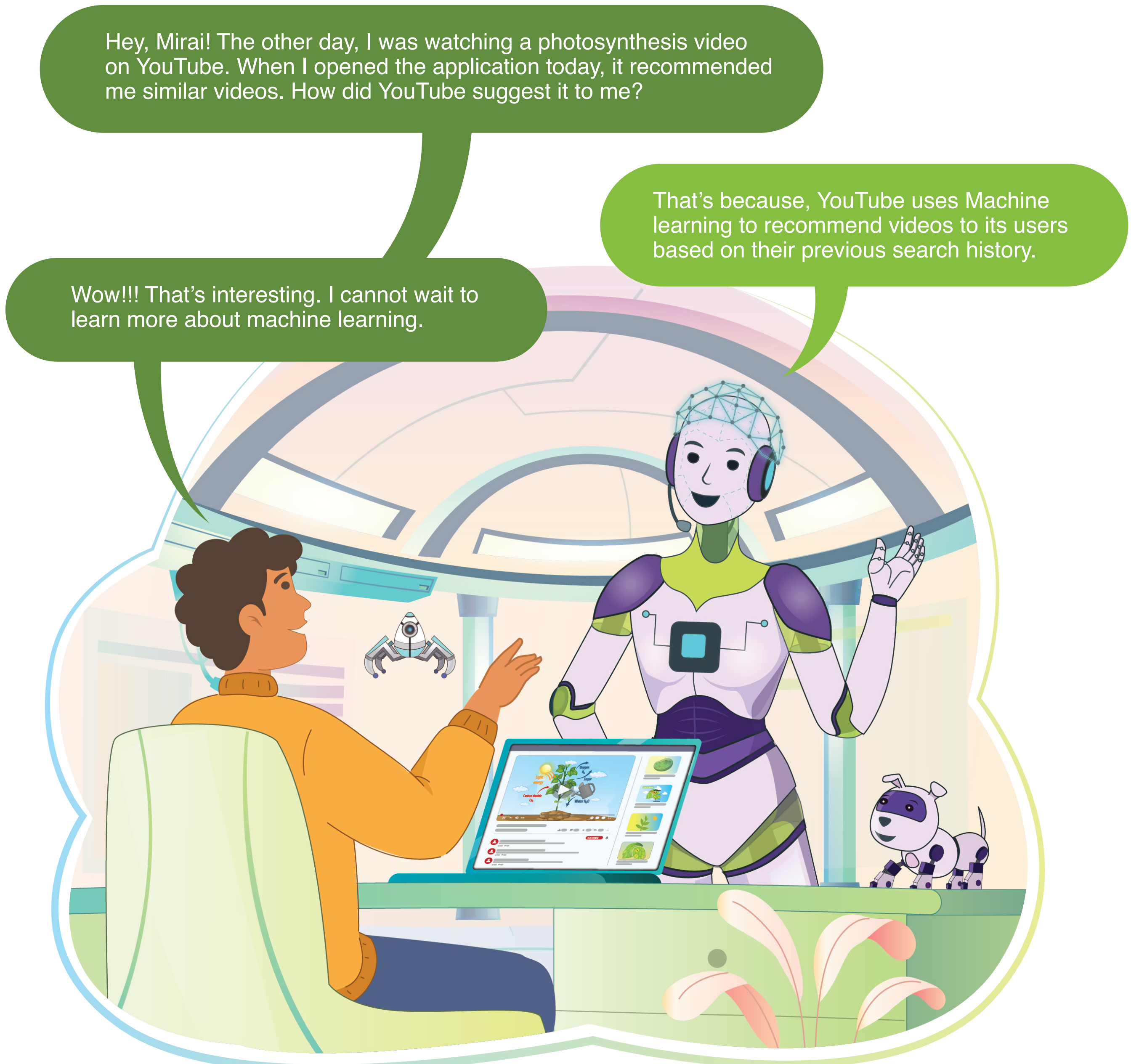


Introduction to Machine Learning

07



7. Introduction to Machine Learning

Machine Learning

- Have you used e-commerce applications on a smartphone to purchase things?
- What did you look for while using the e-commerce platform?

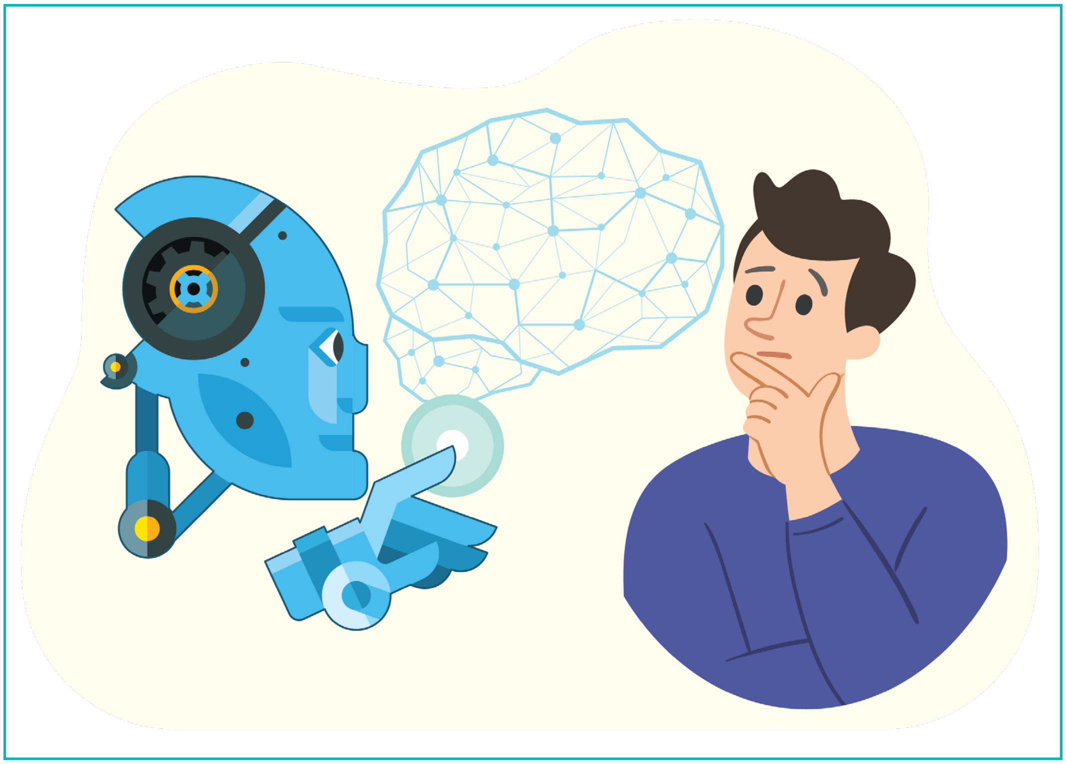
The e-commerce application constantly fetches the searched data and saves our preferences for future use. Although the e-commerce application cannot store every searched item, it does store the most relevant, latest, and frequently searched items.

E-commerce and other similar applications use new programming methods based on training computers. Whatever inputs the user provides, the machine learns from them. With every iteration, the computer learns new information; hence, this approach is called 'Machine Learning'. It is a subset of artificial intelligence.

Let us understand Artificial Intelligence (AI) and its various subcategories.

Artificial Intelligence

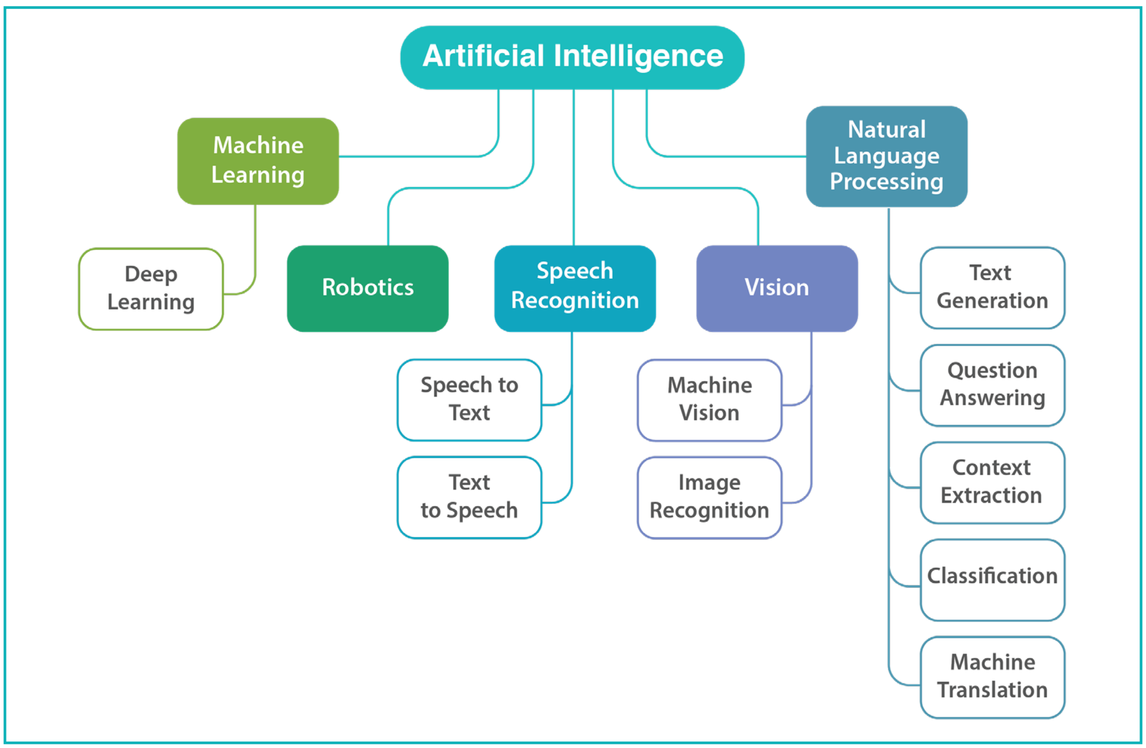
Artificial Intelligence is a method of teaching a computer machine to think and learn as humans do. AI is designed similar to how humans think, learn, decide, and work while solving a problem. The data used to train the computers is called 'training data'. The results of training data are used to build intelligent software systems.



Subsets of Artificial Intelligence

The most common AI subsets are as follows:

- Machine Learning
- Speech Recognition
- Natural Language Processing
- Computer Vision
- Robotics



Subsets of AI

'Machine Learning' is an important component among other subsets of AI.

Exercise

State if the statements are True or False

1. Virtual reality is not a subset of AI.

▲ ______________________

2. Deep learning is a subset of artificial intelligence.

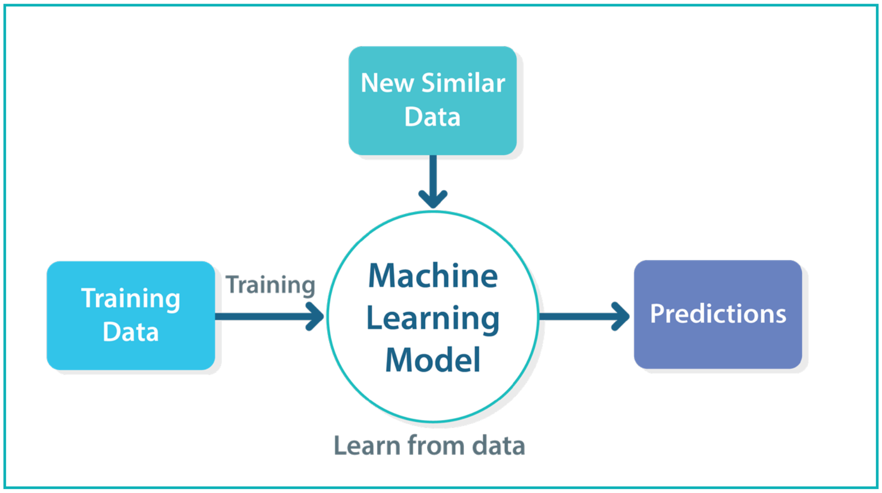
▲ ______________________

3. E-commerce applications store every searched item for future use.

▲ ______________________

Fundamentals of Machine Learning

Machine Learning is a method of data analysis that automates model building. It is a subset of artificial intelligence based on the concept that computers can learn from data, detect correlations, patterns, and make decisions with minimal human intervention. Machine Learning can make predictions autonomously, without being told what to do. It improves the performance of computers through experience.



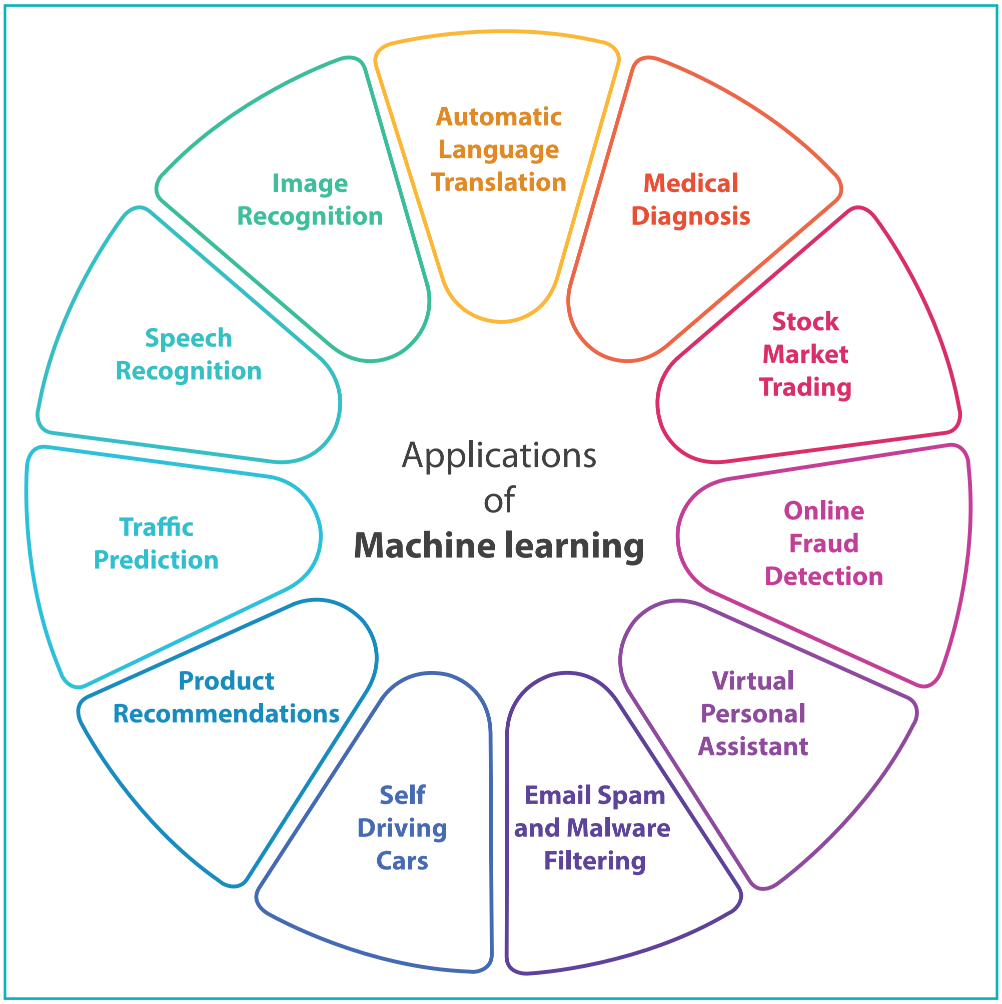
Machine learning model

Relevance and Application of Machine Learning (ML)

Have you ever considered how e-commerce applications can predict which products may interest us?

These applications have a product recommendation engine that analyses user's search results using various Machine Learning algorithms. Based on the results of these algorithms, the engine proposes the most relevant recommendations.

We use Machine Learning in our daily life, often without knowing it, such as Google Maps, Google Assistant, Alexa, etc. Below are some of the most popular real-world applications of Machine Learning:



Applications of machine learning

Bytes

- An Algorithm is a set of instructions for solving a problem or completing a task.
- The Product Recommendation Engine is a filtering system that analyses data from users' input to learn and show the product they want to purchase.

ML algorithms

The Product Recommendation Engine uses a combination of one or more Machine Learning algorithms. Some ML algorithms are linear regression, logistic regression, decision trees, and the k-means algorithm.

Bytes

Regression is a method for estimating the correlation between the variables. For example, 'Y' is dependent on the value of 'X'. Which means, we are trying to understand, how the value of 'Y' changes with respect to change in 'X'.

Linear regression- Linear regression is an algorithm used for estimating the relationship between variables for making predictions.

Hours of study	Percentage marks
8	60 %
10	70 %
14	75 %
12	72 %

Data for Linear Regression

Can you predict the percentage of marks scored by the student studying for 13 hours?

Answer: Yes, it will be either 73 or 74.

The Linear Regression algorithm examines the correlation between the continuous variables. Continuous variables can be defined as the infinite number of values between any two given values. Here, **hours** and **marks** are the continuous variables. Some popular applications of Linear regression are trends and sales analysis, weather forecast, salary prediction, real estate prediction, etc.

Logistic Regression- Logistic Regression algorithm operates on categorical variables. 'Categorical variables' can be defined as a finite number of categories or distinct groups, such as 0 or 1, Yes or No, True or False, Spam or Not Spam, etc. It is a predictive analysis algorithm which works on the concept of probability.

Grade	Hours of study	Result
5	5	Fail
6	6	Fail
7	10	Pass
8	12	Pass
9	11	???

Data for Logistic Regression

Can you predict the result of grade 9?

Answer: Yes, it will be a pass. Here, 'pass' and 'fail' are the categorical variables.

Exercise

Tick mark ☑ the correct answer

4. Which type of regression algorithm has a categorical dependent variable?

   a. Linear regression ☐ b. Logistic regression ☐
   c. Decision tree algorithm ☐ d. K-means algorithm ☐

5. Which of the following is a real-world application of Machine Learning?

   a. Linear regression algorithm ☐ b. Logistic regression algorithm ☐
   c. Product recommendation engine ☐ d. Decision tree algorithm ☐

6. Which of the following is a continuous variable in the context of Linear regression?

   a. Pass or Fail ☐ b. Yes or No ☐
   c. Salary prediction ☐ d. Product recommendation ☐

Fill in the Blanks.

7. ______________________ is a set of instructions for solving a problem.

8. E-commerce applications have ______________________ that analyses user search results using various machine learning algorithms.

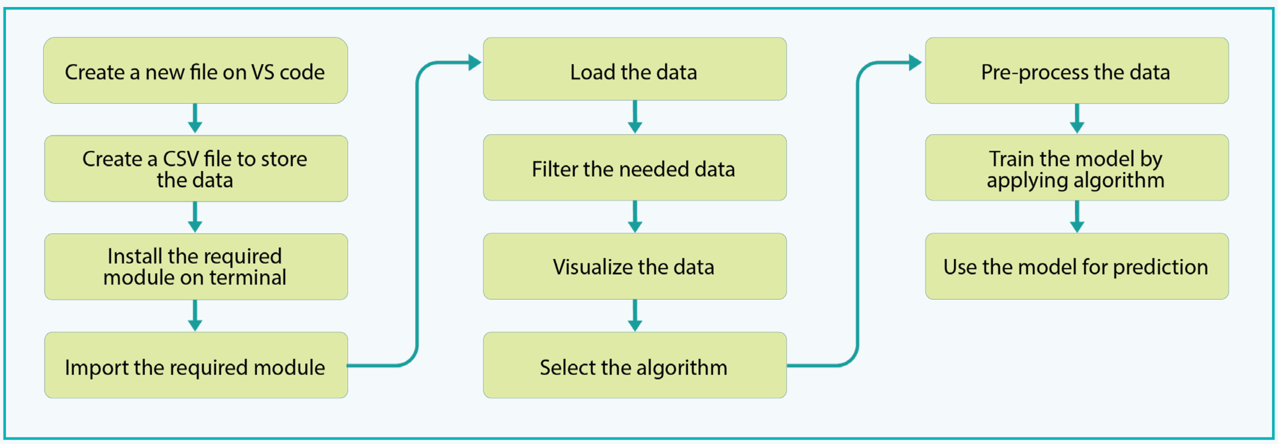
9. Linear and Logistic Regression algorithms are ______________________ analysis algorithms.

10. The more data is fed to the ______________________ application, the more efficient it becomes in decision making.

Activity

Write a Python program for Students Marks Prediction using a Linear Regression Machine Learning algorithm.

The program must predict student marks based on the number of hours studied.



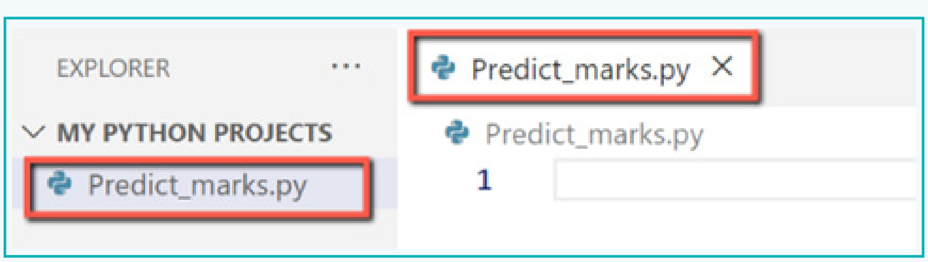
Project flow

1 Create a new file in VS code

a. Create a new folder named “MY PYTHON PROJECTS” in the desired location.

b. Open the VS Code application and create a new workspace by clicking on the File menu and Add Folder to Workspace, then select the folder “MY PYTHON PROJECTS” and click on ‘Add’.

c. Right-click on the folder “MY PYTHON PROJECT” and select “New file”. Give a name to the file as “predict_marks.py”.



Activity

2 Create a CSV File to Store the Data

CSV Files

A **Comma Separated Values (CSV)** file is a plain text file that contains a list of data separated by commas.

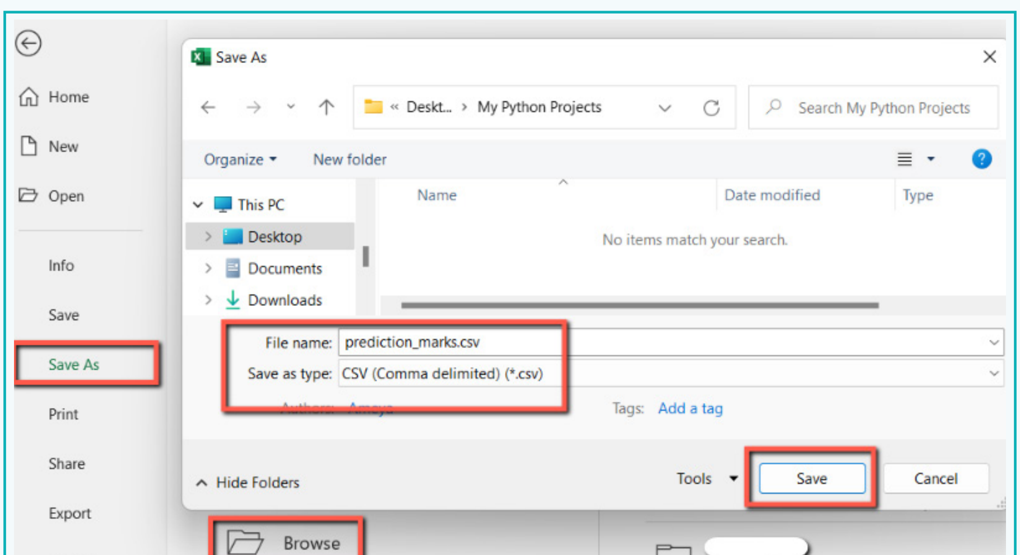
CSV files are easy to use and can be easily imported and exported between different programs. Create a 'CSV' file using the following steps-

a. Open the Ms-Excel application and type the data manually

	A	B
1	Hours	Marks
2	8	60
3	10	70
4	12	72
5	14	75

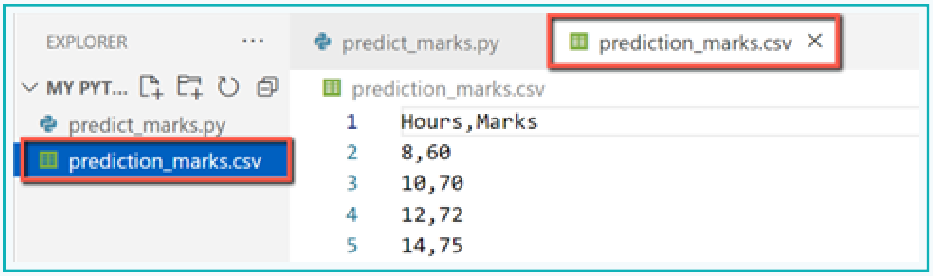
b. Click on 'Save As' to save this file in the 'MY PYTHON PROJECT' folder. Set the file format as 'CSV' from the drop-down list.

c. Now, name it "Prediction_marks" and click on'Save'.



Activity

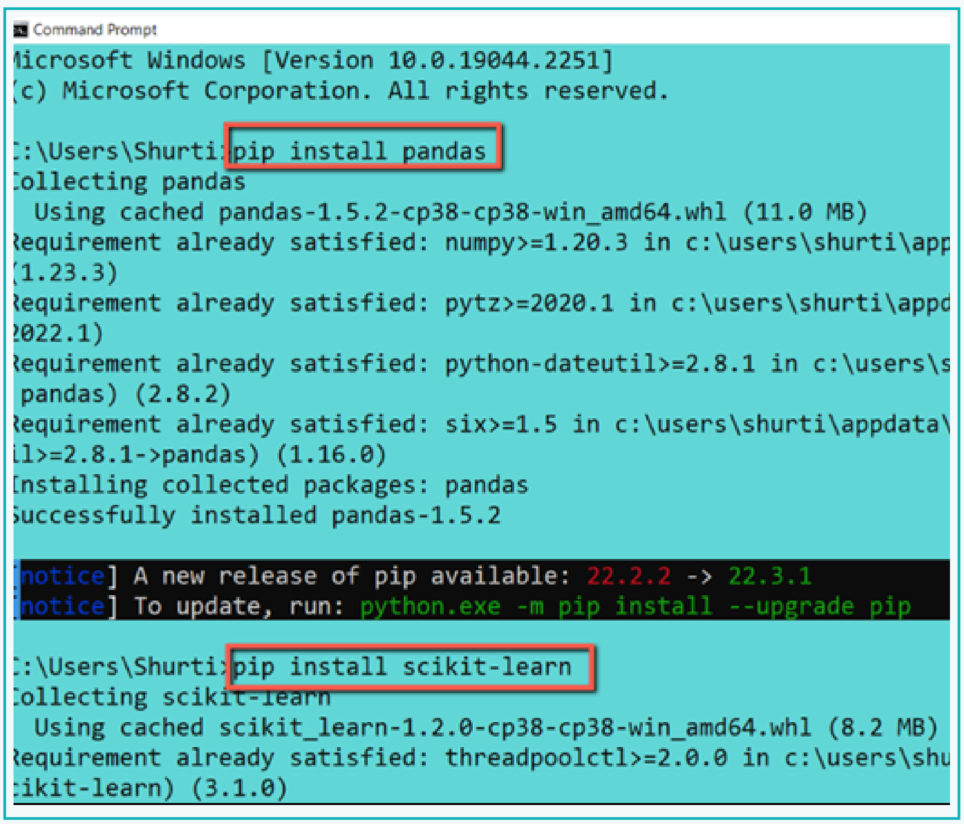
d. Open the VS-code application. The 'Prediction_marks.csv' file will appear here.



3 Install the required modules on the terminal

a. Open the command prompt and type the following pip commands to install pandas, scikit-learn, and matpotlib modules.

- pip install pandas
- pip install scikit-learn
- pip install matplotlib



Activity

```
C:\Users\Shurti>pip install matplotlib
Collecting matplotlib
  Downloading matplotlib-3.6.3-cp38-cp38-win_amd64.whl (7.2 MB)
     ---------------------------------------- 7.2/7.2 MB 1.0 MB/s eta
Requirement already satisfied: python-dateutil>=2.7 in c:\users\shurti\appda
matplotlib) (2.8.2)
Collecting contourpy>=1.0.1
  Downloading contourpy-1.0.7-cp38-cp38-win_amd64.whl (162 kB)
     ---------------------------------------- 163.0/163.0 kB 979.0 kB/s eta
```

4 Import the Required Modules

We require the following modules-

Pandas	It is used for data manipulation and analysis.
NumPy	It is used to perform a wide variety of mathematical operations on arrays. It gets automatically installed when we install the pandas library.
Matplotlib	It is used for data visualization.
Pyplot	It provides an interface for plotting graphs.
Sklearn	It is used as a tool for machine learning and statistical modelling for regression.
sklearn.linear model	It is used for finding out the relationship between the given variables and predicted variables.
linearRegression	It is a class of linear models.

Table 3: Required modules

Activity

a. Import the modules as shown below.

Predict_marks.py > ...

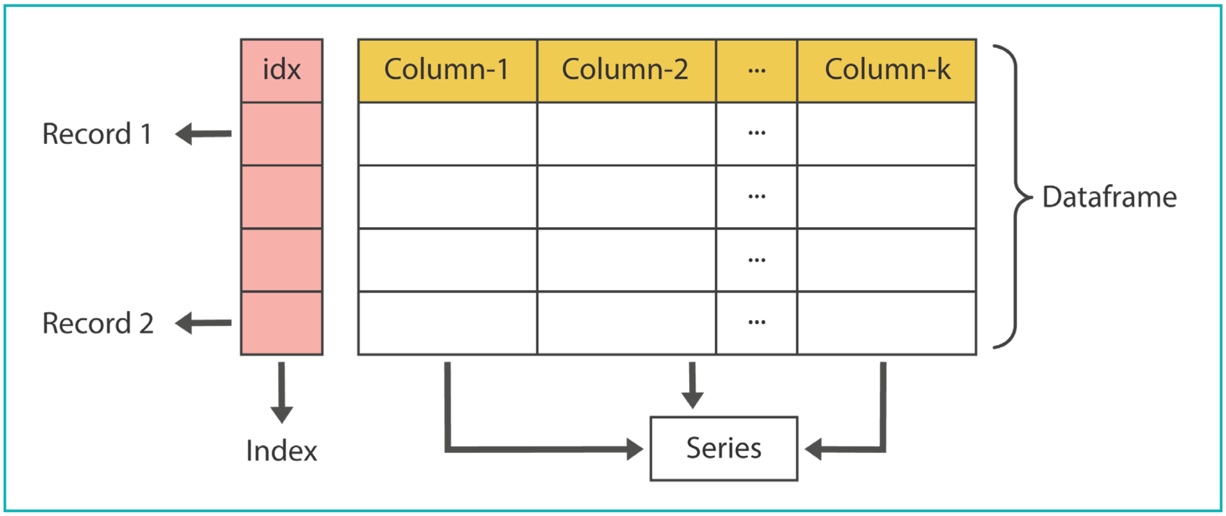
```
#import required modules
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

Load the Data

We load the CSV file data using ‘the read_csv’ function of the ‘pandas’ module and when reading a CSV file with ‘pandas’, the data is saved in a ‘panda DataFrame’. A **panda DataFrame** is a two-dimensional data structure consisting of rows and columns, similar to a spreadsheet.

‘Pandas’ commonly represents data in two ways:

1. **Series** – It is similar to a single column in a spreadsheet that can contain multiple rows, but only one column. Each series is represented as a '1D' array.
2. **DataFrame** - A collection of series that are given an index value by Pandas.



Series and Dataframe

Activity

5 **Load the CSV file 'Prediction_marks' using the 'read_csv' function and save in a variable 'df'. Print the DataFrames on the terminal.**

Syntax: read_csv('filepath')

Note:
Here, we saved the 'prediction_marks' in the same folder.

```

#load your CSV file data as panda's dataframe
df=pd.read_csv("Prediction_marks.csv")
print('\nCSV file is-\n',df)
```

Output:

To see the output, press 'Ctrl+F5'.

```
CSV file is-

   Hours  Marks
0      8     60
1     10     70
2     12     72
3     14     75
```

The 'csv' file is now converted into a 'pandas DataFrame'.

Filter the data

'The indexing operator []' is used to select the required columns. We only need 'Hours' and 'Marks' from the dataframe. To select a single column, simply place the column name between the brackets.

Syntax – DataFrame ['column name']

Activity

6 **Using the 'indexing operator', select two columns from the DataFrame, 'Hours' and 'Marks'. We will train our model using these two columns.**

- Save in the respective variables- 'x' and 'y', and then print them. These 'x' and 'y' are training data.

```
#filter the data by selecting needed column
x=df['Hours']
y=df['Marks']
print('\nneeded dataframes are-\nHours column-\n',x,'\nMarks column-\n',y)
```

Output:

```
needed dataframes are-
Hours column-
 0     8
1    10
2    12
3    14
Name: Hours, dtype: int64 n
Marks column-
 0    60
1    70
2    72
3    75
Name: Marks, dtype: int64
>>> 
```

Activity

Visualize the Data

We use visualization to understand the patterns and relationships in data. For data visualization, we are using the 'pyplot' module of the 'matplotlib' library. The following functions need to be used:

a. **title()** - It is used to specify the title.

   Syntax – title (label, fontsize=None, color, others)

b. **xlabel()** - It is used to set the label for the x-axis.

   Syntax – xlabel (label, fontsize=None, color, others)

c. **ylabel()** - It is used to set the label for the y-axis.

   Syntax – ylabel(label, fontsize=None, color, others)

d. **plot()** - This method is used to create a line plot on the x, y datapoints.

   Syntax – Plot(x, y, data, others)

   x, y: These parameters are the horizontal and vertical coordinates of the data points. The 'x' values are optional.

e. **scatter()** - This method is used to draw a scatter plot. A 'scatter plot' is a diagram where each value in the data set is represented by a 'dot'.

   Syntax – Scatter(x-axis, y-axis, color, linestyle, edgecolor, others)

f. **grid()** - It is used to add grid lines to the plot of points on x, y.

   Syntax – grid(color, axis, linestyle, linewidth, others)

g. **show()** - It is used to display all figures. We cannot see the effects of all the functions without calling this function.

   Syntax - show()

Activity

7 **Set the title as “Marks prediction”, ‘x’ label as ‘Hours’ and the ‘y’ label as ‘Marks’.**

```
#Data visualization
#set title and labels for graph
plt.title('Marks prediction',color='b',fontsize=16)
plt.xlabel('Hours',color='b',fontsize=14)
plt.ylabel('Marks',color='b',fontsize=14)
```

a. Plot the graph on the ‘x’ and ‘y’ axes.

b. Draw the scatter points on x and y axes with blue dots and a solid line.

c. The ‘grid()’ method will add grids on both axes.

d. The ‘show()’ function will display all the effects used in the preceding steps, such as labels, titles, grids, and so on.

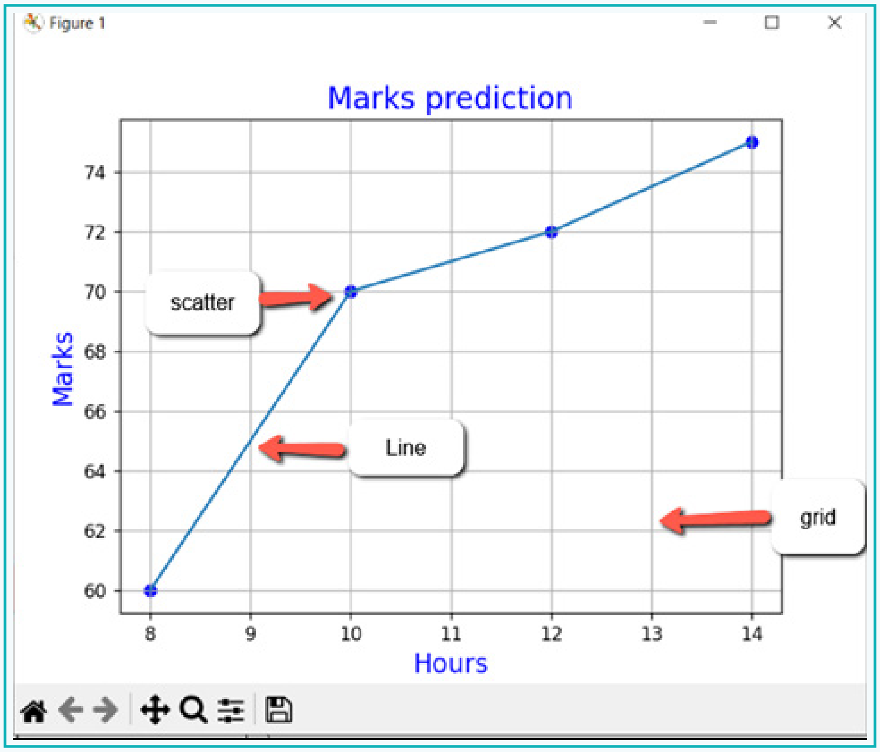
```
#plot the graph
plt.plot(df.Hours,df.Marks)

#plot scatter points
plt.scatter(df.Hours,df.Marks,color='b',linestyle='solid')

#make grids and show the complete graph
plt.grid()
plt.show()
```

Activity

Output:



Select the Algorithm

Here, we will use the basic ML algorithm— 'linear regression'. We have already imported the linear regression class from the sklearn's linear model library.

8 **Create an object for a linear regression class and name it as "model".**

```

#introduce the model
#create object of linear regression
model=LinearRegression()
```

Activity

Pre-process the Data

In the next step, we need to use the ‘fit()’ method of the linear regression class that will take two parameters ‘x’ and ‘y’ (training data) in which ‘x’ should be a 2D array. For this, we need to first use two methods-

1. **‘to_numpy’()-** Convert the data in the ‘x’ column or series into 1D array.

   Syntax- Pandas series.to_numpy()

2. **reshape()-** Convert 1D array into 2D array.

   Syntax- Pandas series.reshape(r,c)

   r= Number of rows in the resultant array.

   c= Number of columns in the resultant array.

9 **Convert the ‘Hours’ column that is ‘x’ into 1D array using the ‘to_numpy ’method and save it in the same variable ‘x’.**

a. Print the 1D array of the ‘Hours’ series to the terminal.

```
#preprocess the data
#convert x and y series into array
x=x.to_numpy()
print('\narray of Hours series-',x)
```

Output:

```
array of Hours series- [ 8 10 12 14]
```

Activity

b. Change the shape of 1D array into 2D array using the 'reshape' method.

Syntax - x.reshape(-1,1)

Here,

Rows = -1; when rows are unknown, they are represented by '–1'.

Column = 1; Here, column must be one so that the linear regression model knows that these values are for the 'hours' column only.

Print the reshaped array.

```
#reshape the array as single feature
x=x.reshape(-1,1)
print('\nafter reshape of Hours array-\n',x)
```

Output:

```
after reshape of Hours array-
 [[ 8]
 [10]
 [12]
 [14]]
```

10 Train the Model by Applying Algorithm

Now, using the 'fit()' method of the linear regression class, we will fit our data in the Linear regression model.

Syntax- fit (training data)

Here, 'x' and 'y' are the training data.
x: must be a 2D array
y: must be a 1D array (pandas series)

Activity

- Fit the data in the linear regression model using the 'fit' method. 'x' and 'y' are the arguments.

```
#fit the data in model
model.fit(x,y)
```

11 Use the Model for Prediction

Predict the model using the 'predict' method.

Syntax- Model.predict (x)

Where, x: Must be a 2D numpy array.

We want to predict for a value 13.

a. Convert this value into a 1D array → [13]

b. Then, we reshape (-1,1) required for model understanding → [[13]]

c. Therefore write model.predict([[13]])

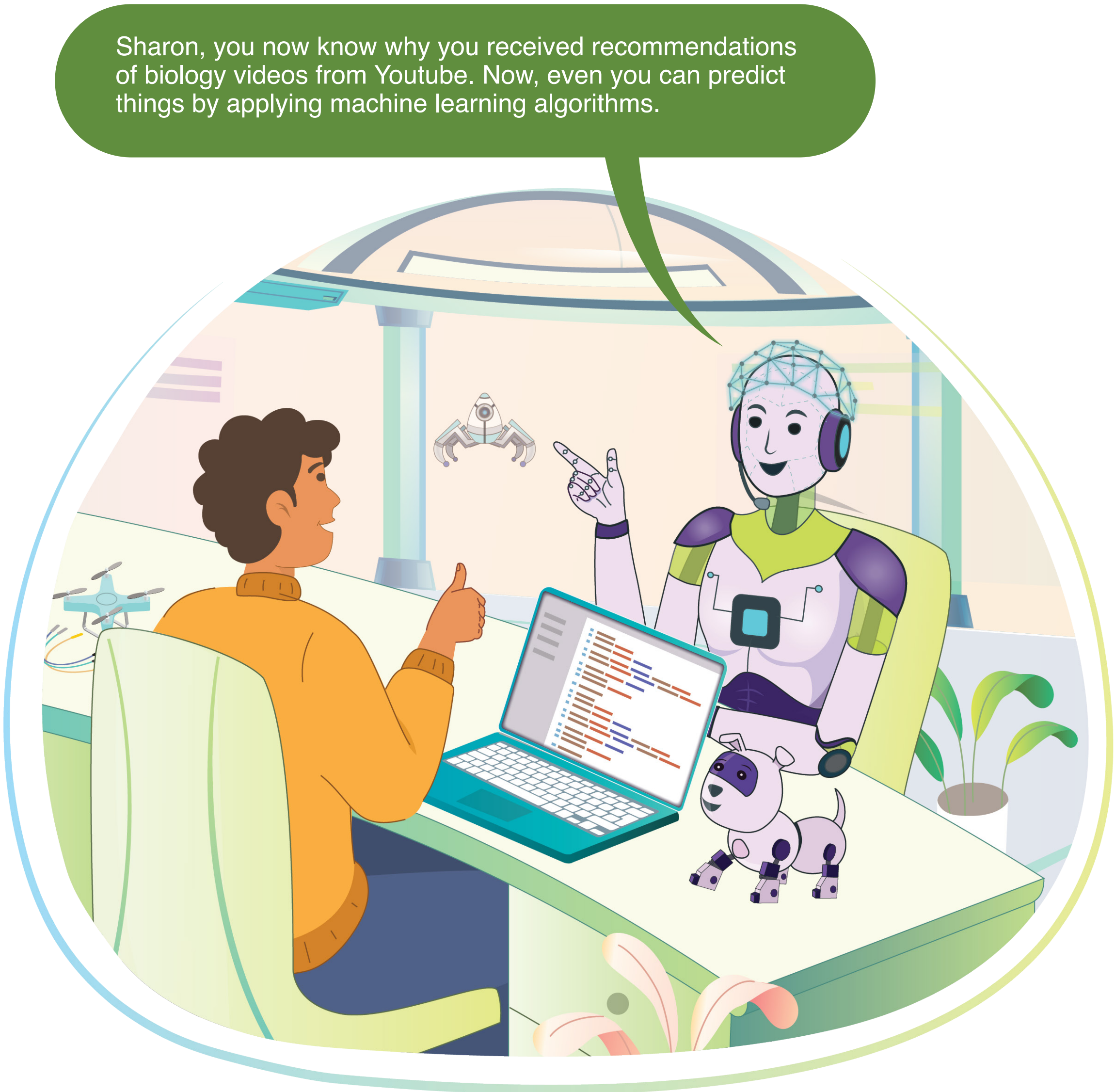
d. Print the predicted result.

```
#predcit the model
Predicted_marks=model.predict([[13]])
print("\n Predicted marks are",'=',int(Predicted_marks))
```

Final Output

```
Predicted marks are = 73
```

In this lesson, we have studied about Artificial Intelligence and its subsets. We have explored the fundamentals of Machine Learning and its algorithms. We learnt about the Linear Regression algorithm and implemented it to predict the marks scored based on the number of hours studied.



Tech Flash

- **Recommender Engine** : It is software or a special-purpose program that uses algorithms to analyse the information at hand and make useful recommendations.
- **Deep Learning** : The subset of Machine Learning in which neural networks with a large data set can create powerful machine learning models.
- **Data Pre-processing** : It is the initial step in every machine learning project. It involves cleaning, transforming, and organising raw data so that it can be further analysed. It ensures that the data is accurate, consistent, and in the required format for analysis by ML algorithms.